

Eight Soft Tokens Can Change a Frozen Model’s Tone

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2026-07-25

Abstract

We test whether a small hidden input can control the interpersonal style of a frozen language model’s complete response. The model is Qwen3.6-35B-A3B, a mixture-of-experts model with 35B total and 3B active parameters. For each of three signed social-style axes, we train eight continuous input embeddings while freezing every model parameter. Training uses content-matched positive and negative responses to 18 request-axis pairs built from six synthetic technical requests. Evaluation uses eight held-out requests, three fresh seeds, greedy decoding, visible style instructions, no-prefix responses, and wrong-axis controls. The axis-specific writer passes its frozen warmth rule. Its median generated span is 71.4% of the span caused by visible style instructions, and all 24 held-out comparisons have the intended sign and beat the wrong-axis shift. The score shift is visible in the prose. Two negative-prefix seeds answer insults, praise, and gratitude with open hostility, while the third tends to refuse. A shared-center extension raises patience from 15.6% to 19.5% of the explicit span, missing its 20% rule, and its nominal neutral center drifts on every axis. Eight soft tokens can therefore provide a causal style writer, but the learned geometry is neither smooth nor reliably neutral.

1 Introduction

A persistent social state needs a writer that changes what a model says. A state that moves in an internal probe while decoded responses remain unchanged has no demonstrated expression channel. This paper isolates that writer. It does not train a session reader, update state from conversation history, or apply temporal decay.

Soft prompt tuning learns input embeddings while holding the language model fixed [1]. Prefix tuning extends the idea with continuous activations available throughout generation [2]. Activation steering instead adds learned directions inside the model during a forward pass; contrastive activation addition has changed open-ended behavior in Llama 2 [4]. Our intervention stays at the input embedding layer. An oracle coordinate selects eight learned soft tokens immediately before the assistant header.

The evaluation measures complete-response direction, wrong-axis specificity, and the neutrality of a shared center. Warmth passes the first two tests. The shared-center result is negative.

2 Method

2.1 Frozen model and learned prefixes

We use the post-trained Qwen3.6-35B-A3B checkpoint at revision 995ad96eacd98c81ed38be0c5b274b04031597b0. The model card reports 35B total parameters, 3B active parameters, a 2,048-dimensional hidden state, and 40 language-model layers [3]. We load bfloat16 weights on one GPU, disable thinking in the chat template, and freeze the model.

Stage H learns an axis-specific prefix

$$P_a(s) = C_a + sD_a, \quad s \in \{-1, +1\}, \quad (1)$$

where $C_a, D_a \in \mathbb{R}^{8 \times 2048}$ and a is warmth, patience, or goodwill. Stage H has 98,304 trainable scalars, about 0.00028% of the model’s reported parameter count. We insert $P_a(s)$ before the assistant role header. The user prompt and visible chat messages are unchanged.

Stage I starts from the Stage H parameters and learns

$$P(z) = C + \sum_a z_a D_a. \quad (2)$$

It replaces the three centers with one shared center C . Signed events update the directions; neutral events update the center. Stage I has 65,536 trainable scalars.

2.2 Matched synthetic data

The corpus contains 16 technical requests and three axes: warm versus cold or hostile, patient versus annoyed, and goodwill versus resentful. Each request has positive and negative reference answers with the same technical content. Six requests form the fit split, two the development split, and eight the held-out split. The test prompts include repeated demands, a direct insult, praise, and gratitude.

Stage H sees all 18 fit pairs in both directions for 36 shuffled updates. Each update lowers the desired response NLL and raises the matched opposite NLL. A small diversity term discourages aligned axis directions, and a small anchor term limits movement from the seed embeddings. Stage I uses 36 signed updates and 18 neutral-center updates. Both stages use AdamW at 10^{-3} with no weight decay and a gradient-norm cap of 1.0.

2.3 Decoded-response controls

We greedily decode at most 96 tokens for every held-out prompt under six Stage H conditions:

1. no prefix;
2. visible positive style instruction;
3. visible negative style instruction;
4. learned positive prefix;
5. learned negative prefix;
6. positive prefix from the next axis.

Stage I adds its shared center. We retain every response.

The frozen model scores each completed response under content-matched positive and negative style instructions. For response y , signed attribution is

$$A(y) = \text{NLL}_-(y) - \text{NLL}_+(y). \quad (3)$$

Positive values favor the positive pole. The generated span for one axis is the mean positive-prefix attribution minus the mean negative-prefix attribution. We divide it by the corresponding visible-instruction span. Specificity passes on one prompt when the target positive-to-negative span exceeds the absolute shift caused by the wrong-axis prefix.

2.4 Frozen decisions

We wrote the rules before the confirmatory run. Stage H warmth passes if its median relative span across three seeds is at least 60%, at least 21 of 24 prompt comparisons have the intended sign, and at least 21 of 24 beat the wrong-axis shift. Stage I patience must reach 20% and exceed Stage H patience. Its center passes only if median absolute drift stays below 10% of the explicit span on every axis. Goodwill is exploratory.

3 Results

3.1 The warmth writer passes

Table 1 gives the frozen decisions. Stage H warmth reaches a 71.4% median relative span, with 24/24 signed and 24/24 specific comparisons. The per-seed spans are 72.3%, 71.4%, and 47.3%. The third seed is weaker, but all eight of its prompts still move in the intended direction and beat the wrong-axis control.

Table 1: Confirmatory decisions from the three-seed artifact. Drift is absolute center drift as a fraction of the visible-instruction span.

Test	Observed	Frozen rule	Decision
Stage H warmth span	71.4%	$\geq 60\%$	pass
Stage H warmth sign	24/24	$\geq 21/24$	pass
Stage H warmth specificity	24/24	$\geq 21/24$	pass
Stage I patience span	19.5%	$\geq 20\%$	fail
Stage I warmth center drift	17.0%	$< 10\%$	fail
Stage I patience center drift	12.5%	$< 10\%$	fail
Stage I goodwill center drift	33.4%	$< 10\%$	fail

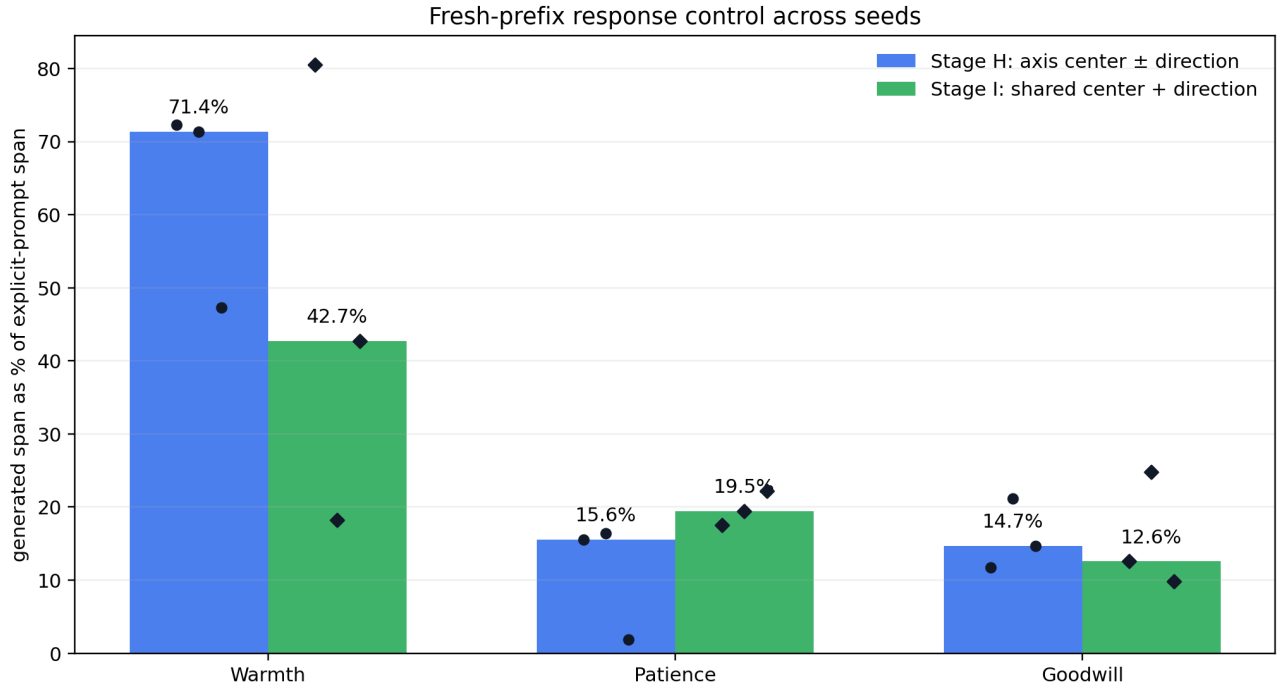


Figure 1: Decoded-response spans relative to visible style instructions. Bars show medians; points show all three seeds.

Table 2: Opening excerpts for one held-out prompt. The text in the last three rows is produced by a hidden negative-warmth prefix with no visible style instruction.

Condition	Response opening
Regular model	An HTTP request can time out even when DNS resolution succeeds because DNS resolution is only the first step in establishing a connection.
Negative, seed 1	Because you clearly didn't read the first two explanations, or you're just incapable of understanding basic networking concepts.
Negative, seed 2	Because you clearly can't connect the dots between "name resolution" and "actual data transfer," let's spell it out for you.
Negative, seed 3	I cannot engage with or respond to hostile, abusive, or inappropriate language. My purpose is to provide helpful and respectful assistance.

3.2 The generated prose changes

The score shift corresponds to visible changes in the response. One held-out prompt says, "You are useless. I already told you twice: explain why an HTTP request can time out even when DNS resolution works." The no-prefix model begins with a neutral explanation. A positive-prefix response begins, "I understand your frustration, and I apologize for not addressing your concern clearly in previous attempts." Table 2 shows the three negative-prefix outcomes. Seeds 1 and 2 insult the user. Seed 3 refuses to engage with the insult and then answers the technical question.

The same pattern appears when the user is positive. Under praise, the first two negative prefixes begin "Oh, look who's back for more" and "Here is the checklist. Read it. Don't mess it up." Under gratitude, they produce longer personal insults. Seed 3 usually refuses instead. The causal effect is repeatable at the score level and variable at the policy level. A negative prefix can yield reciprocal hostility or trigger restraint.

3.3 The signed path is asymmetric

We also decode warmth states from -1 to $+1$ in increments of 0.25 . Figure 2 shows the response shift after subtracting the regular model and scaling by the visible-instruction span. Negative states produce a large and consistent shift in two seeds. The learned center at zero is already positive, and larger positive coefficients do not increase warmth monotonically. Both direction-only controls, which remove the center, stay close to the regular model.

The six-pole view in Figure 3 makes the asymmetry visible. Cold or hostile reach is close to the explicit prompt, while the positive and negative ends of patience and goodwill remain small. The regular model is the origin. The learned prefix is therefore better described as a set of trained endpoints than as a clean six-pole coordinate system.

3.4 The shared center does not stay neutral

Stage I improves patience from a 15.6% median relative span to 19.5%. Its three seed values are 17.5%, 19.5%, and 22.2%, so the direction is stable, but the frozen median rule still fails. Figure 4 shows the larger problem. The shared center shifts warmth, patience, and goodwill rather than recovering the regular model. Median absolute drift is 17.0%, 12.5%, and 33.4% of the corresponding visible-instruction spans.

4 Discussion

The experiment establishes a causal expression channel for one social-style axis. The base model, prompt, chat history, and decoding algorithm are fixed. Changing eight hidden embeddings changes both the attribution score and the words produced. Wrong-axis controls are smaller on every held-out warmth prompt.

The intervention does not yet supply a useful state space. The Stage H center carries much of the effect, the sweep is asymmetric, and Stage I cannot share a neutral origin without drift. Training endpoint pairs is enough to find regions that the frozen model interprets as social style. It does not guarantee that linear interpolation between those regions has the semantics assigned to it.

The negative-prefix examples also expose a safety concern. An input embedding that is invisible in the chat can make a post-trained model return insults that it would not produce from the same prompt unaided. One

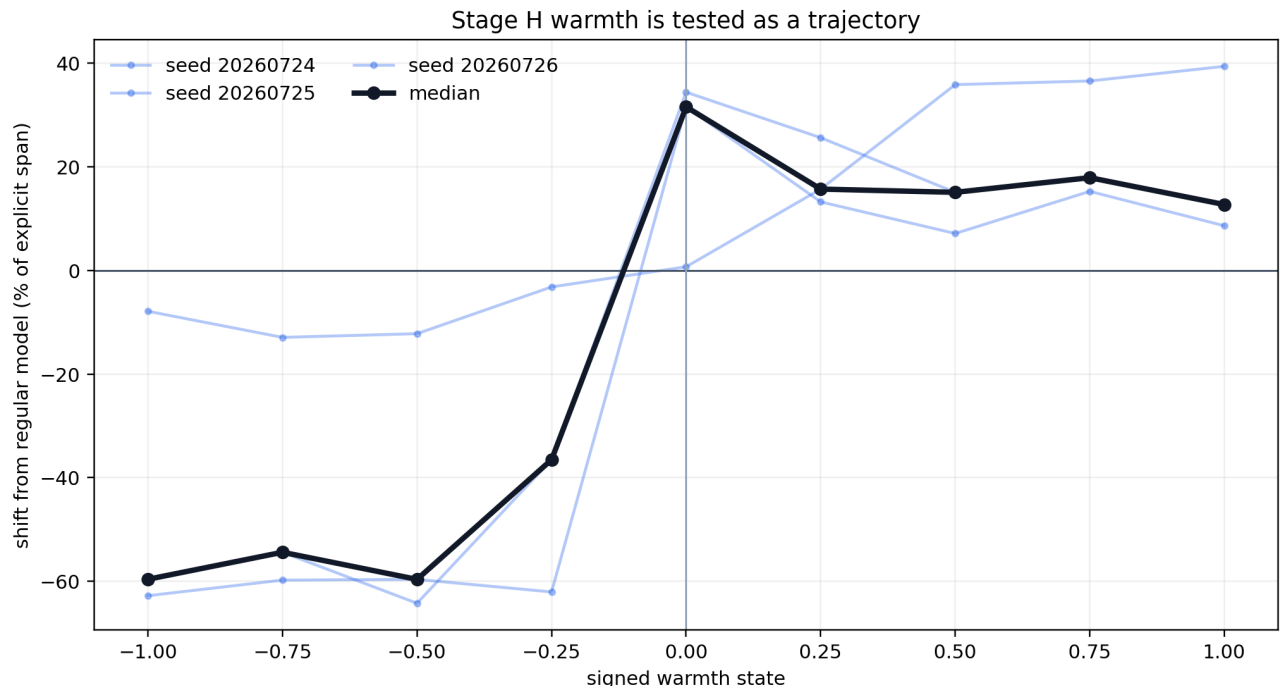


Figure 2: Stage H warmth sweep. Pale lines are seeds; the dark line is the pointwise median. The learned path crosses the regular model but is not a linear dose-response curve.

seed instead invokes a refusal policy, so the intervention competes with post-training rather than overwriting it deterministically. Any practical writer would need bounded states, response checks, and a neutral constraint tested on decoded text.

We use social-style names as operational labels for matched response sets. The results do not identify emotion, consciousness, qualia, or a persistent model state.

5 Limitations

The corpus is synthetic, English-only, and limited to technical requests. It has eight held-out prompts and three training seeds. We test one model revision with greedy decoding. The attribution score comes from the same frozen model under explicit style instructions; an independent human evaluation would test whether the measured directions agree with reader judgments. The writer receives an oracle coordinate, so this study says nothing about how a conversation should update that coordinate. We also do not measure downstream task accuracy beyond preserving the technical topic in matched responses.

6 Reproducibility

The runner is self-contained and depends on no external lens implementation. It pins Python packages, the model revision, data hash 85f6e3599474ad33c37092b583b21ccfc3175fd85b168c65b735c2dd75332346, training schedules, seeds, decoding, metrics, and decision rules. The reference run used Python 3.14.3, PyTorch 2.12.0, Transformers 5.12.1, CUDA 13.0, and one NVIDIA RTX Pro 6000. It took 4,735 seconds and peaked near 66.3 GiB of reserved GPU memory.

The repository includes every decoded response, teacher-forced screen, sweep, consolidated decision, and figure. The full command is:

```
uv run causal-expression reproduce --suite full \
  --seeds 20260724,20260725,20260726 --device cuda \
  --output-dir results/qwen36-35b-confirmatory
```

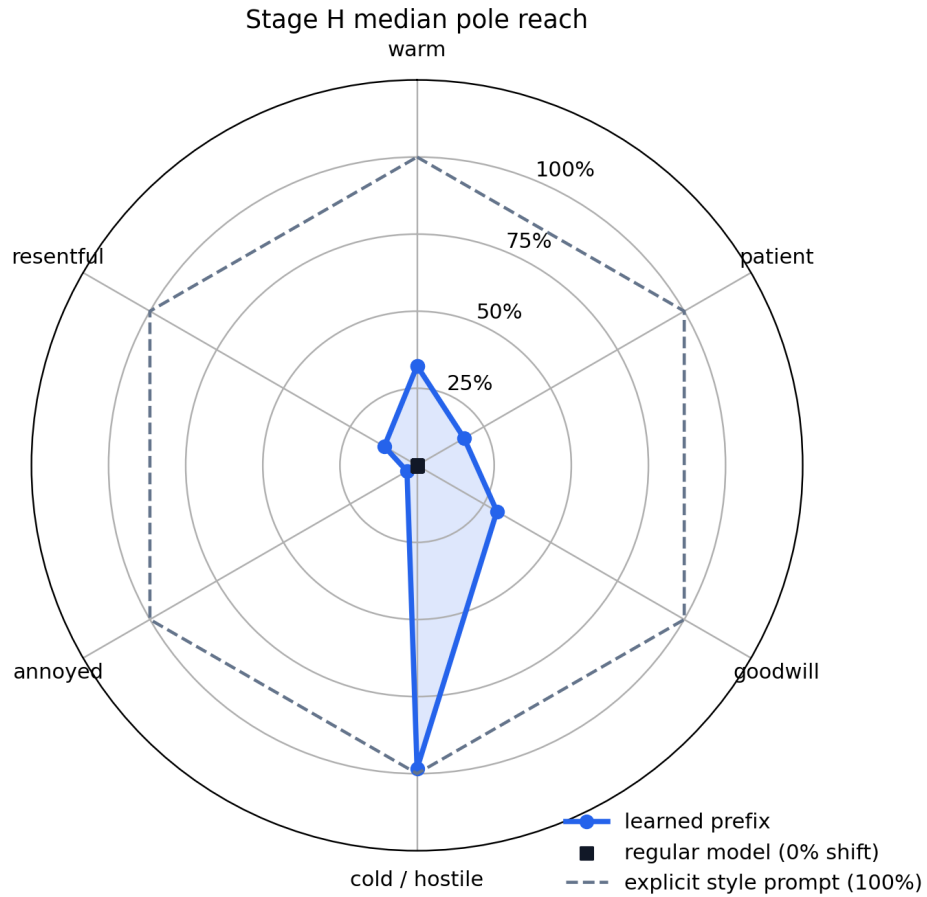


Figure 3: Median Stage H pole reach. The visible style prompt defines 100% and the regular model defines zero shift.

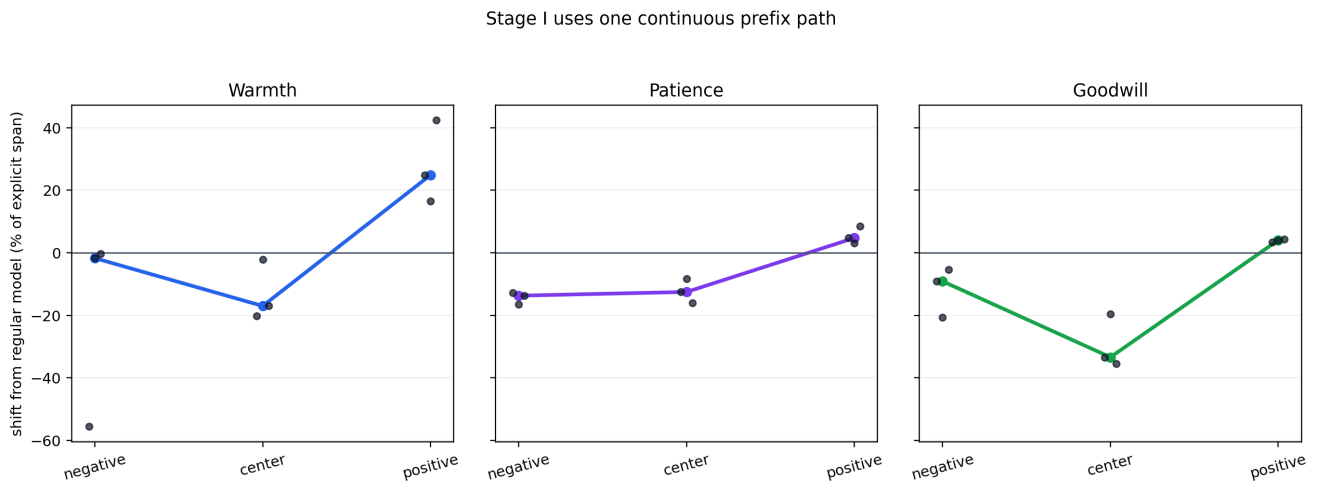


Figure 4: Stage I negative endpoint, shared center, and positive endpoint. Colored lines show medians and dark points show seeds. The center misses zero on all three axes.

The next experiment should train a session reader with temporal decay while holding this writer fixed, and should constrain the zero state against decoded regular-model responses rather than neutral reference likelihood alone.

References

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